

9) Take the Institution name as input. Use Pydantic to define the schema for the desired output and create a custom output parser. Invoke the Chain and Fetch Results. Extract the below Institution related details from Wikipedia: **The founder of the Institution. When it was founded. The current branches in the institution . How many employees are working in it. A brief 4-line summary of the institution.**

```
# Install required libraries
```

```
!pip install wikipedia-api pydantic
```

```
import re
import wikipediaapi
from pydantic import BaseModel, Field
from typing import List, Optional

# --- Data Schema ---
class InstitutionDetails(BaseModel):
    name: str
    founder: Optional[str] = None
    founded: Optional[str] = None
    branches: List[str] = Field(default_factory=list)
    number_of_employees: Optional[int] = None
    summary: Optional[str] = None
```

```
# --- Extraction Logic ---
```

```
def fetch_institution_details(institution_name: str) -> InstitutionDetails:
    # 1. Setup Wikipedia API with a proper User-Agent
    user_agent = "InstitutionScraper/1.0 (contact: myemail@example.com)"
    wiki = wikipediaapi.Wikipedia(user_agent=user_agent, language='en')
    page = wiki.page(institution_name)

    if not page.exists():
        raise ValueError(f"The page for '{institution_name}' does not exist.")

    full_text = page.text
```

2. Helper function to extract info using Regex

```
def extract_pattern(pattern, text, is_list=False):
    match = re.search(pattern, text, re.IGNORECASE)
    if match:
        content = match.group(1).strip()
        if is_list:
            return [item.strip() for item in content.split(',')]
        return content
    return [] if is_list else None
```

3 Extraction logic

```
founder_pattern = r"(?:founded|established|started)\s+by\s+([\^.\n,]+)"
```

```
founder_match = re.search(founder_pattern, full_text, re.IGNORECASE)
```

```
founder = founder_match.group(1).strip() if founder_match else "Unknown"
```

```
year_pattern =
```

```
r"(?:founded|established|started|incorporated)(?:\s+in)?(?:\s+the\s+year)?\s+(\d{4})"
```

```
year_match = re.search(year_pattern, full_text, re.IGNORECASE)
```

```
founded = year_match.group(1) if year_match else "Unknown"
```

```
branches = extract_pattern(r"Branches\s*[:\~]?s*(.*)", full_text, is_list=True)
```

```
raw_employees = extract_pattern(r"Number of employees\s*[:\~]?s*([\d,]+)", full_text)
```

```
emp_count = None
```

```
if raw_employees:
```

```
    try:
```

```
        emp_count = int(raw_employees.replace(',', ''))
```

```
    except ValueError:
```

```
        emp_count = None
```

```
return InstitutionDetails(
```

```
    name=page.title,
```

```
    founder=founder,
```

```
    founded=founded,
```

```
    branches=branches,
```

```
    number_of_employees=emp_count,
```

```
    summary=page.summary[:500] + "..."
```

```
)
```

```
# --- Execution ---
try:
    val=input("Enter the Institution name:")
    data = fetch_institution_details(val)
    print(data.model_dump_json(indent=2))
except Exception as e:
    print(f"Error: {e}")
```

OUTPUT:

```
... Enter the Institution name: MIT

{
  "name": "Massachusetts Institute of Technology",
  "founder": "Salvador Luria with a National Cancer Institute
grant and staffed by a cohort recruited with David Baltimore",
  "founded": "1861",
  "branches": [],
  "number_of_employees": null,
  "summary": "The Massachusetts Institute of Technology (MIT)
is a private research university in Cambridge, Massachusetts,
United States. Founded in 1861 to advance \"useful
knowledge\", the university has played a significant role in
the development of many areas of technology and
science.\nWilliam Barton Rogers founded MIT to accelerate
American industrialization through scientific knowledge.
Initially funded by a federal land grant, the institute
adopted a German polytechnic model emphasizing laboratory
inst..."
}
```